

# Chapter 1: Systems of linear equations and Gauss elimination

## Section 1.1: Introduction

### 1 An application for systems of linear equations (question)

**Situation:** A chemist mixes 8 litres of liquid A with 4 litres of liquid B. The resulting mixture contains precisely 30 grams of sodium chloride (this can be determined, for example, by evaporating the water and analysing the residue). When adding 4 litres of A to 6 litres of B, a total of 29 grams sodium chloride is found in the mixture.

**Question:** What is the sodium chloride concentration (in grams per litre) of liquids A and B?

### 2 An application for systems of linear equations (solution)

Write  $a$  for concentration in liquid A, in g/l (grams per litre).

Write  $b$  for concentration in liquid B, in g/l.

Then  $x$  litres of liquid A contain  $x \cdot a$  grams sodium chloride,

and  $x$  litres of liquid B contain  $x \cdot b$  grams sodium chloride.

The text gives us the following information:

$$\left. \begin{array}{rcl} 8a & + & 4b = 30 \\ 4a & + & 6b = 29 \end{array} \right\}$$

This is a *system of linear equations*. We solve it by subtracting half of the first equation from the second.

$$(II) - \frac{1}{2}(I) : \left. \begin{array}{rcl} 8a & + & 4b = 30 \\ & & 4b = 14 \end{array} \right\}$$

Thus  $b = 14/4 = 3.5$  and  $a = (30 - 4b)/8 = 2$ .

**Answer:** Liquid A contains 2g/l sodium chloride, and liquid B contains 3.5g/l.

### 3 What are linear equations

We are interested in *linear equations* where each variable occurs only in the form “number times variable”. Examples in the two variables  $x$  and  $y$ :

|                       |   |                                          |
|-----------------------|---|------------------------------------------|
| $2x - 8y = 8$         | ✓ | linear                                   |
| $3x^2 + 8y = 9$       | ✗ | not linear since $x^2$ appears           |
| $3x + 8\sin(y) = 9$   | ✗ | not linear since $\sin(y)$ appears       |
| $9^2x + \sin(9)y = 7$ | ✓ | linear: $\sin(9)$ and $9^2$ are numbers! |

### 4 Solutions of equations

Finding *solutions* of an equation means: find values for the variables so that the *equations* become *equalities* when the values are substituted for the variables.

---

For example, the equation

$$x^2 - 3y = 1$$

has solution  $x = 2$ ,  $y = 1$ . To check, just substitute the values for the variables:

$$x^2 - 3y = 2^2 - 3 \cdot 1 = 1 \quad \checkmark$$

It *also* has solution  $x = 4$ ,  $y = 5$ , and the solution  $x = 8$ ,  $y = 21$ .

---

An equation may have *several* solutions.

## 5 Systems of linear equations

Systems of linear equations are “simultaneous equations”.

$$\text{Example: } \left. \begin{array}{rcl} 2x_1 & + & 3x_2 = 6 \\ 4x_1 & + & 4x_2 = 4 \end{array} \right\} \quad (1)$$

Task: *find all pairs of real numbers  $(x_1, x_2)$  which satisfy both of these equations at the same time.*

One solution is given by  $x_1 = -3, x_2 = 4$ :

$$\left. \begin{array}{rcl} 2 \cdot (-3) & + & 3 \cdot 4 = -6 + 12 = 6 \quad \checkmark \\ 4 \cdot (-3) & + & 4 \cdot 4 = -12 + 16 = 4 \quad \checkmark \end{array} \right\}$$

---

### Questions:

- How can we find such a solution?
- Is this the only solution? *How many* solutions are there?
- Can we “list” all solutions?

## 6 A systems of linear equations may have no solution

$$\left. \begin{array}{rcl} 2x_1 & + & 3x_2 = 6 \\ 4x_1 & + & 6x_2 = 4 \end{array} \right\}$$

*This system has no solution.* Reason: If  $2x_1 + 3x_2 = 6$  then

$$4x_1 + 6x_2 = 2 \cdot \underbrace{(2x_1 + 3x_2)}_{=6} = 2 \cdot 6 \neq 4$$

so we *cannot* satisfy both equations at the same time.

---

A system of linear equations may have no solution at all.

## 7 A systems of linear equations may have a single solution only

$$\left. \begin{array}{rcl} 2x_1 & + & 3x_2 = 6 \\ 0x_1 & - & 2x_2 = -8 \end{array} \right\}$$

Can can determine a solution by “backward substitution”, that is, “from bottom to top”:

$$x_2 = \frac{-8}{-2} = 4$$

$$x_1 = \frac{1}{2}(6 - 3x_2) = \frac{1}{2}(6 - 12) = -3$$

There is no choice: *there is precisely one solution.*

---

A system of linear equations may have precisely one solution.

## 8 A systems of linear equations may have infinitely many solutions

$$\left. \begin{array}{rcl} x_1 & + & x_2 - x_3 = 4 \\ & 2x_2 & + 2x_3 + 3x_4 = 8 \\ & & 2x_4 = 4 \end{array} \right\}$$

Can can determine a solution by “backward substitution”:

$$x_4 = \frac{4}{2} = 2$$

$x_3$  arbitrary value – a “parameter”

$$x_2 = \frac{1}{2}(8 - 2x_3 - 3x_4) = 1 - x_3$$

$$x_1 = 4 - x_2 + x_3 = 3 + 2x_3$$

There are infinitely many solutions: *choose* any value for the parameter  $x_3$ , and *calculate* the value of the other variables.

A system of linear equations may have infinitely many solutions.

## 9 A homogeneous system always has the trivial solution

If a system of linear equations is *homogeneous* (that is, all the right-hand sides are 0), then “all variables zero” is a solution. This is sometimes called the *trivial solution*.

**Example:** The homogeneous system of linear equations

$$\left. \begin{array}{rcl} 2x_1 & + & 3x_2 = 0 \\ -x_1 & + & 5x_2 = 0 \end{array} \right\}$$

has the solution  $x_1 = 0, x_2 = 0$ .

- A homogeneous system of linear equations always has the trivial (all-zero) solution.
- It may have other solutions as well.

## 10 Summary: the number of solutions of a system of linear equations

A system of linear equations may have

- *no solution* at all, or

- a *unique solution* solution (that is, precisely one solution), or
- *infinitely many* solutions.

We will see soon that *there are no other possibilities*.

---

A *homogeneous* system of linear equations always has the trivial solution.

## Section 1.2: Row echelon form

### 11 The general system of linear equations

A system with  $m$  linear equations in  $n$  variables looks like this:

$$\left. \begin{array}{cccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & b_m \end{array} \right\} \quad (\text{S})$$

The numbers  $a_{i,j}$  are the *coefficients*, the  $x_j$  are the *variables*, and the  $b_i$  are given real numbers.

We say informally that  $x_j$  is a *later* variable than  $x_i$  if  $j > i$ . That is,  $x_2$  is later than  $x_1$ , and  $x_3$  is later than  $x_2$  and  $x_1$ .

### 12 Row echelon form: Introduction

If each successive equation in a system starts with a later variable than the previous equation, we can read off how many (and what) solutions there are.

$$\left. \begin{array}{ccc} 2x_1 & + & 3x_2 & = & 6 \\ & -2x_2 & & = & -8 \end{array} \right\}$$

(Exactly one solution, can calculate  $x_2$  and then  $x_1$ )

---

$$\left. \begin{array}{rclcl} x_1 & + & x_2 & - & x_3 & = & 4 \\ & & 2x_2 & + & 2x_3 & + & 3x_4 & = & 8 \\ & & & & & & 2x_4 & = & 4 \end{array} \right\}$$

(Infinitely many solutions, can choose any value for  $x_3$ )

---

$$\left. \begin{array}{rclcl} x_1 & + & x_2 & - & x_3 & = & 4 \\ & & 2x_2 & + & 2x_3 & + & 3x_4 & = & 8 \\ & & & & & & 0 & = & 0 \\ & & & & & & 0 & = & 3 \end{array} \right\}$$

(No solutions, last equation never satisfied!)

## 13 Row echelon form: Definition

A system of linear equations is in *row echelon form* if the following conditions hold:

- **(RE1)** If the left-hand side of an equation is identically zero (all variables have coefficient 0), then the same is true for all equations below it.
- **(RE2)** If the left-most non-zero coefficient in some equation is the coefficient of the  $\ell$ th variable  $x_\ell$ , then in every equation below it the coefficients of  $x_1, x_2, \dots, x_\ell$  are all 0.

Informally:

- If no variable whatsoever occurs in some equation (left-hand side is 0), the same must be true for all equations below it.
- Each successive equation must start with a later variable than the previous one.

## 14 Row echelon form: Examples

$$\left. \begin{array}{rclclcl} 3x_1 & + & 6x_2 & + & x_3 & - & 5x_4 & = & b_1 \\ & & x_2 & & & & & = & b_2 \\ & & & & & & 7x_4 & = & b_3 \\ & & & & & & 0 & = & b_4 \\ & & & & & & 0 & = & b_5 \end{array} \right\} \quad (\text{E})$$

System (E) is in row echelon form:  $x_3$  is a parameter, while  $x_1$ ,  $x_2$  and  $x_4$  are not.

$$\left. \begin{array}{rclcl} 4x_2 & + & 8x_3 & = & b_1 \\ 5x_2 & + & x_3 & = & b_2 \\ & & 0 & = & b_3 \\ & & 2x_3 & = & b_4 \end{array} \right\} \quad (\text{N})$$

System (N) is **not** in row echelon form: condition (RE2) is violated in second equation, and (RE1) is violated in fourth equation.

## 15 Row echelon form: reading off solutions

We can read off the solution from a system in row echelon form:

- All equations of the form  $0 = 0$  can be ignored. They are satisfied no matter what value the variables take.
- If the system contains an equation of the form  $0 = b$  with  $b \neq 0$ , the system has *no solution*.
- Otherwise solutions exist; they can be read off by “backward substitution”.
  - If  $x_j$  is the earliest variable (with non-zero coefficient) in some equation, then  $x_j$  can be expressed in terms of the later variables  $x_{j+1}, x_{j+2}, \dots, x_n$ .
  - If  $x_j$  is never the earliest variable in an equation, then  $x_j$  is a parameter which is allowed to take on any value.



Suppose there exists at least one solution.

- If there is a parameter (that is, a variable that is not the “first” variable in any equation) then there are infinitely many solutions.
- If there is no parameter, then there is a unique solution.

## 16 Strategy to solve systems of linear equations

**Idea:** Given any system of linear equations, produce a new system of linear equations that

- has the same solutions as the given system, and
- is in row echelon form.

Once we have achieved this, we can read off easily how many solutions there are, and we can determine the solutions explicitly.

The procedure we use is called **Gauss elimination**, and will be explained shortly.

## 17 Solving systems of linear equations: Example 1

$$\left. \begin{array}{rcl} 2x_1 & + & 3x_2 = 6 \\ 4x_1 & + & 4x_2 = 4 \end{array} \right\} \quad (\text{A})$$

Subtracting the double of first equation from second gives a new system *with the same solutions*:

$$(II) - 2(I) \quad \left. \begin{array}{rcl} 2x_1 & + & 3x_2 = 6 \\ 0x_1 & -2x_2 = & -8 \end{array} \right\} \quad (\text{B})$$

Why same solution? Clearly every solution of (A) is a solution of (B). But we can get system (A) from (B) by *adding* the double of the first equation to the second, so the argument works backwards too!

As system (B) is in row echelon form, we can read off that there is a unique solution:  $x_2 = 4$  and  $x_1 = -3$ .

## 18 Systems of linear equations: Example 2

An example with three variables and three equations, together with transformations that bring it into row echelon form:

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ 2x_1 & + & x_2 & + & 4x_3 & = & 3 \\ -4x_1 & - & 2x_2 & - & 5x_3 & = & -4 \end{array} \right\}$$

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ & & & & 3x_3 & = & 2 \\ & & & & -3x_3 & = & -2 \end{array} \right\} \begin{array}{l} \text{(II)-(I)} \\ \text{(III)+2(I)} \end{array}$$

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ & & & & 3x_3 & = & 2 \\ & & & & 0 & = & 0 \end{array} \right\} \begin{array}{l} \\ \text{(III)+(II)} \end{array}$$

Can now read off solutions: parameter  $x_2$ , so infinitely many solutions:  $x_2 \in \mathbb{R}$  arbitrary,  $x_3 = \frac{2}{3}$ , and  $x_1 = \frac{1}{6} - \frac{x_2}{2}$ .

## 19 Systems of linear equations: Example 3

Consider the system with left-hand side as before but different right-hand side. **We use the exact same transformations to get to row echelon form:**

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ 2x_1 & + & x_2 & + & 4x_3 & = & 3 \\ -4x_1 & - & 2x_2 & - & 5x_3 & = & 4 \end{array} \right\}$$

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ & & & & 3x_3 & = & 2 \\ & & & & -3x_3 & = & 6 \end{array} \right\} \begin{array}{l} \text{(II)-(I)} \\ \text{(III)+2(I)} \end{array}$$

$$(III)+(II) \quad \left. \begin{array}{rclcl} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ & & & & 3x_3 & = & 2 \\ & & & & 0 & = & 8 \end{array} \right\}$$

There is no solution as the last equation  $0 = 8$  is never satisfied!

## Section 1.3: Gauss elimination

### 20 Operations in systems of linear equations

Consider a (general) system of linear equations:

$$\left. \begin{array}{rclcl} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & b_m \end{array} \right\}$$

The solutions of the system *do not change* if we apply any of the following three operations:

- **(R1)** Swap two equations.
- **(R2)** Multiply an equation with a non-zero real number.
- **(R3)** Add a multiple of one equation to another equation.

### 21 Gauss elimination: input data and idea

We want to solve the a system of  $m$  linear equations in  $n$  variables:

$$\left. \begin{array}{rclcl} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & b_m \end{array} \right\} \quad (S)$$

We will do so by transforming the system (S) into row echelon form by systematically applying operations (R1), (R2) and (R3). This process is called “Gauss elimination”.

## 22 Gauss elimination: the procedure

**Step 0:** If all the coefficients  $a_{i,j}$  are zero, skip to reading off the solutions.

**Step 1: Swap rows.** Apply (R1) so that the left-most non-zero coefficient occurs in the first equation of the system. Let’s say this is the coefficient of  $x_j$ , and has value  $t$ .

**Step 1.5 (optional): Divide.** Apply (R2) to the first equation; multiply it with  $1/t$ . Now  $x_j$  appears with coefficient 1 in the first equation.

**Step 2: Subtract.** Apply (R3) to all equations **below** the first; subtract from the  $k$ th equation a multiple of the first so that in the new  $k$ th equation  $x_j$  has coefficient 0.

**Repeat:** Now you repeat steps 1, 1.5 and 2 with the system obtained by “forgetting” the first equation, and the variables  $x_1, x_2, \dots, x_j$ ; then you repeat again “forgetting” another equation and more variables, *etc.*; finally, the system will be in row echelon form.

## 23 Gauss elimination: an explicit example

Consider the following system of three equations in five variables  $x_1, \dots, x_5$  — the real number  $b$  is a constant:

$$\left. \begin{array}{rrrrr} x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 6x_2 & + & 6x_3 & - & 6x_4 & + & 10x_5 & = & b \end{array} \right\}$$

Where’s  $x_1$ ? It doesn’t appear (has coefficient 0 in all equations), so we can rewrite this like so:

$$\left. \begin{array}{rrrrr} 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 6x_2 & + & 6x_3 & - & 6x_4 & + & 10x_5 & = & b \end{array} \right\}$$

## 24 Gauss elimination: an explicit example (continued 1)

So far we have:

$$\left. \begin{array}{rrrrrr} 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 6x_2 & + & 6x_3 & - & 6x_4 & + & 10x_5 & = & b \end{array} \right\}$$

$x_2$  is first variable to (really) appear anywhere.

It doesn't appear in first equation, but in second. So perform a "Step 1": swap first and second equation.

$$\left. \begin{array}{l} (II) \\ (I) \end{array} \begin{array}{rrrrrr} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 0x_1 & + & 6x_2 & + & 6x_3 & - & 6x_4 & + & 10x_5 & = & b \end{array} \right\}$$

## 25 Gauss elimination: an explicit example (continued 2)

$$\left. \begin{array}{rrrrrr} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 0x_1 & + & 6x_2 & + & 6x_3 & - & 6x_4 & + & 10x_5 & = & b \end{array} \right\}$$

Have 0 as coefficient for  $x_2$  in second equation already. Need to deal with third one. Perform a "Step 2": subtract first from third.

$$\left. \begin{array}{l} \\ \\ (III) - (I) \end{array} \begin{array}{rrrrrr} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 0x_1 & + & 0x_2 & + & 2x_3 & - & 4x_4 & + & 6x_5 & = & b - 8 \end{array} \right\}$$

## 26 Gauss elimination: an explicit example (continued 3)

$$\left. \begin{array}{rrrrrr} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 0x_1 & + & 0x_2 & + & 2x_3 & - & 4x_4 & + & 6x_5 & = & b - 8 \end{array} \right\}$$

Now we “forget” first equation, and variables  $x_1$  and  $x_2$  (don’t touch those again!). Need to make  $x_3$  disappear below second equation. Perform a “Step 2”: Subtract double of second equation from third.

$$(III) - 2(II) \left. \begin{array}{rclclcl} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ & & & & & & 0 & = & b - 10 \end{array} \right\}$$

## 27 Gauss elimination: an explicit example (continued 4)

We have arrived at a system of linear equations in row echelon form:

$$\left. \begin{array}{rclclcl} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ & & & & & & 0 & = & b - 10 \end{array} \right\}$$

Now read off solutions.

Last equation is satisfied if and only if  $b = 10$ . Thus:

There is **no** solution if  $b \neq 10$ .

## 28 Gauss elimination: an explicit example (continued 5)

$$\left. \begin{array}{rclclcl} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ & & & & & & 0 & = & b - 10 \end{array} \right\}$$

If  $b = 10$  then last equation is just  $0 = 0$  which can be ignored. Since only  $x_2$  and  $x_3$  appear as left-most variables with non-zero coefficient somewhere in the system **the variables  $x_1$ ,  $x_4$  and  $x_5$  are parameters**.

## 29 Gauss elimination: an explicit example (continued 6)

$$\left. \begin{array}{rrrrrr} 0x_1 & + & 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 0x_1 & + & 0x_2 & + & x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ & & & & & & 0 & = & b - 10 \end{array} \right\}$$

Still assuming  $b = 10$ , we get from second equation

$$x_3 = 1 + 2x_4 - 3x_5 ,$$

and from first equation

$$\begin{aligned} 6x_2 &= 8 - 4x_3 + 2x_4 - 4x_5 \\ &= 8 - 4(1 + 2x_4 - 3x_5) + 2x_4 - 4x_5 \\ &= 4 - 6x_4 + 8x_5 \end{aligned}$$

## 30 Gauss elimination: an explicit example (summary)

Solving the system of linear equations in five(!) variables  $x_1, \dots, x_5$  (the real number  $b$  being a constant)

$$\left. \begin{array}{rrrrr} x_3 & - & 2x_4 & + & 3x_5 & = & 1 \\ 6x_2 & + & 4x_3 & - & 2x_4 & + & 4x_5 & = & 8 \\ 6x_2 & + & 6x_3 & - & 6x_4 & + & 10x_5 & = & b \end{array} \right\}$$

yields this result:

If  $b \neq 10$ : no solution.

If  $b = 10$ : infinitely many solutions. Variables  $x_1, x_4$  and  $x_5$  are parameters (arbitrary value), and

$$\begin{aligned} x_2 &= \frac{1}{3}(2 - 3x_4 + 4x_5) , \\ x_3 &= 1 + 2x_4 - 3x_5 . \end{aligned}$$

### 31 Matrix of coefficients (MOC)

It is notationally easier to retain information on coefficients only: The homogeneous system

$$\left. \begin{array}{cccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & 0 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & 0 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & 0 \end{array} \right\}$$

can be reconstructed from its “matrix of coefficients” (MOC):

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{pmatrix}$$

**Note:** The MOC sees *only* the coefficients, but *not* the variables!

### 32 Augmented matrix of coefficients (AMOC)

Similarly, the system

$$\left. \begin{array}{cccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & b_m \end{array} \right\}$$

can be reconstructed from its “augmented matrix of coefficients” (AMOC):

$$\left( \begin{array}{cccc|c} a_{1,1} & a_{1,2} & \dots & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} & b_m \end{array} \right)$$

**Note:** The AMOC sees *only* the coefficients and right-hand sides, but *not* the variables!



### 33 Row operation on matrices

Corresponding to the operations on systems of linear equations, we have the following “row operations”:

- **(R1)** Swap two rows of the matrix.
- **(R2)** Multiply a row with a non-zero real number.
- **(R3)** Add a multiple of one row to another row.

We also have a notion of row echelon form, and we can perform Gauss elimination with matrices only.

### 34 Row echelon form for matrices

The important thing to understand is that *row echelon form is a condition on the matrix of coefficients, not the augmented matrix*. This is because the right-hand sides of the equations are irrelevant when it comes to row echelon form.

So a matrix of coefficients (not augmented!) is in row echelon form if

- below each all-zero row are only all-zero rows, and otherwise
- each successive row starts with more zeros.

An augmented matrix of coefficients is in row echelon form if it is so after forgetting the last column (the “augmented” part that corresponds to the right-hand sides of the equations).

### 35 Example: row echelon form for matrices

Here is an example of an AMOC in row echelon form:

$$\left( \begin{array}{cccc|c} 3 & 2 & 0 & 4 & 2 \\ 0 & 0 & 2 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{array} \right)$$


---

And here is an example of an AMOC that is *not* in row echelon form:

$$\left( \begin{array}{cccc|c} 0 & 2 & 0 & 4 & 2 \\ 1 & 0 & 2 & 6 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 3 & 2 & -2 \end{array} \right)$$

Indeed, the second row starts with fewer zeros than the first, and the fourth row is not all-zero on the left even though the row above is.

### 36 Example: Gauss with matrices

Solve the following system ( $a$  and  $b$  given constants):

$$\left. \begin{array}{rrcr} 2x_1 & + & x_2 & + & 3x_3 & = & -1 \\ 2x_1 & + & x_2 & + & x_3 & = & -a \\ 6x_1 & + & 3x_2 & + & 13x_3 & = & b-4 \end{array} \right\}$$

Start Gauss elimination on AMOC:

$$\begin{aligned} \left( \begin{array}{ccc|c} 2 & 1 & 3 & -1 \\ 2 & 1 & 1 & -a \\ 6 & 3 & 13 & b-4 \end{array} \right) & \xrightarrow[(III)-3 \cdot (I)]{(II)-(I)} \left( \begin{array}{ccc|c} 2 & 1 & 3 & -1 \\ 0 & 0 & -2 & -a+1 \\ 0 & 0 & 4 & b-1 \end{array} \right) \\ & \xrightarrow{(III)+2 \cdot (II)} \left( \begin{array}{ccc|c} 2 & 1 & 3 & -1 \\ 0 & 0 & -2 & -a+1 \\ 0 & 0 & 0 & b-2a+1 \end{array} \right) \end{aligned}$$

### 37 Example: Gauss with matrices (continued)

The last AMOC corresponds to a system in row echelon form:

$$\left. \begin{array}{rrcr} 2x_1 & + & x_2 & + & 3x_3 & = & -1 \\ & & & & -2x_3 & = & -a+1 \\ & & & & 0 & = & b-2a+1 \end{array} \right\}$$

As before, can read off solutions:

- If  $b - 2a + 1 \neq 0$ : no solution.
- If  $b - 2a + 1 = 0$ , we have  $x_3 = \frac{1}{2}(a - 1)$  and

$$x_1 = \frac{1}{2}(-1 - x_2 - 3x_3) = \frac{1}{2}\left(\frac{1}{2} - x_2 - \frac{3}{2}a\right) = \frac{1 - 3a - 2x_2}{4}$$

with  $x_2$  a parameter (infinitely many solutions!).

## 38 Summary

- **Any system of linear equations can be solved explicitly** by transforming it into row echelon form using Gauss elimination.
- We can read off solutions from a system in row echelon form by backward substitution.
- **We obtain structural information** from Gauss elimination: Does a solution *exist* at all? If so, *how many* solutions are there?
- A system of linear equations has
  - either no solution at all,
  - or precisely one solution (*viz.*, if there is a solution but there is no *parameter*),
  - or infinitely many solutions (*viz.*, if there is a solution and there is at least one *parameter*).
  - Systems of linear equations can be treated in matrix form,

which is notationally easier than a full system with variables.